

Mofe Omolade

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EDUCATION

University of British Columbia, *BASc in Computer Engineering*

Expected 2029

Relevant Coursework: Circuit Analysis, Computing Systems, Software Construction, Linear Algebra, Embedded C.

EXPERIENCE

FulcrumAir, *Electronics Engineering Intern*, Calgary, AB

May 2026 - Present

- Engineered compact power distribution networks, CAN/RS-485 communication interfaces, and motor control PCBA architecture for 24V industrial robotics platforms, prioritizing reliable operation on 240kV energized power lines.
- Executed the design and layout of an 8-layer motor control PCBA in Altium around Teensy 4.1, IFX007T, MAX33040E, and LMR50410-Q1, optimizing layer stackup, thermal regulation, and power integrity.
- Developed C++ firmware for multi-channel closed-loop motor control, parsing Hall-effect sensor data at 1 kHz to achieve <0.2 mm linear actuator position accuracy.
- Integrated a power hold-up circuit and interrupt-driven C++ firmware to safeguard telemetry and write critical status registers to EEPROM within 5 ms of detecting power loss during sudden supply cutoffs.

University of Calgary, *Electrical Engineering Research Assistant*, Calgary, AB

July 2023 - August 2023

- Modelled and simulated circuits in LTspice to support research on novel Op-Amp-based oscillator topologies.
- Validated circuit performance, achieving >95% correlation between simulation and lab measurements.
- Utilized oscilloscopes, signal generators, and DMMs to debug test circuits and verify frequency responses.

UBC Rover, *Electrical Subteam*, Vancouver, BC

September 2025 - Present

- Co-designed power architecture, telemetry communication, and motor driver boards for an autonomous test rover.
- Tested a powertrain prototype around the L298N full-bridge motor driver and Nucleo F446RE MCU.
- Developed low-level C firmware using STM32 HAL for real-time telemetry, command parsing, and Hall effect encoder motor feedback at 115200 baud UART.

PROJECTS

Self-Balancing Two-Wheel Robot

- Designing a two-wheeled self-balancing robot using an ESP32-C3, MPU-6050 IMU, and L298N motor driver.
- Developed C++ I2C drivers to parse IMU registers at 1kHz and apply complementary filtering, reducing drift by 85%.
- Implemented a closed-loop PID control system to drive dual DC motors to maintain dynamic equilibrium.

Speedometer Heads-Up Display

- Designed a low-latency heads-up display using an Arduino Nano, GY-NEO6MV2 module, and SPI OLED display.
- Wrote C++ firmware to parse 10 Hz NMEA data streams to compute velocity within 2% of OBD-II metrics.
- Integrated a USB-C-to-DIN step-down regulator to safely isolate vehicle transient spikes up to 18V.

Non-Isolated Bidirectional Synchronous Buck-Boost Converter

- Simulated a bidirectional DC-DC converter in LTspice to achieve 94% peak efficiency across 12V and 24V rails.
- Tuned closed-loop PID compensation in MATLAB and C, keeping voltage ripple within $\pm 1.5\%$ under dynamic loads.
- Designed a 2-layer PCB in Altium with optimized switching loops to minimize radiated EMI and thermal dissipation.

SKILLS

- Hardware: PCBA Design, Power Electronics, CAN, SPI, I2C, UART, RS-485, STM32, ESP32, Arduino.
- Tools: Altium, KiCAD, LTspice, MATLAB, Git, VS Code, STM32 CubeIDE.
- Languages: C/C++, Python, Verilog, Java.
- Lab: Oscilloscopes, Logic Analyzers, DMM, Function Generators, Soldering, Circuit Debugging.